

JEE-Main-30-01-2024 (Memory Based)

[MORNING SHIFT]

Physics

Question: A particle of mass 'm' has been thrown at an angle of 30° with horizontal at speed 'u'. Find its angular momentum at the highest point about the point of projection.

Options:

- (a) $\frac{mu^2}{16g}$
- (b) $\frac{\sqrt{3}mu^2}{16g}$
- (c) $\frac{mu^3}{8g}$
- (d) $\frac{\sqrt{3}mu^3}{8g}$

Answer: (b)

Question: The work function of metal surface is 3 eV. The maximum wavelength that should strike the metal for emitting Photoelectron (in nm) is

Options:

- (a) 413
- (b) 450
- (c) 315
- (d) 350

Answer: (a)

Question: If the length of a rod is doubled and area of cross section is halved, then its young's modulus will be

Options:

- (a) Doubled
- (b) Same
- (c) Halved
- (d) Four times

Answer: (b)

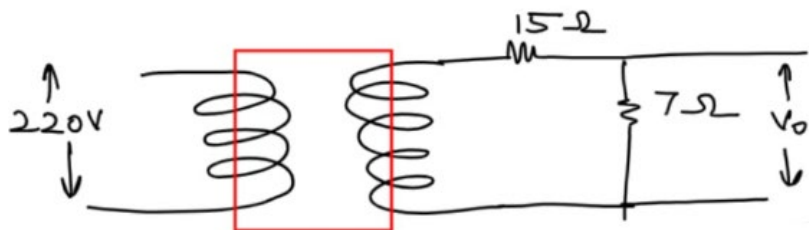
Question: The gravitational potential at certain height from the surface of earth is -5.12×10^7 and gravity at that height is 6.4 ms^{-2} . Then that height from surface of earth is

Options:

- (a) 80 km
- (b) 8000 m
- (c) 800 km
- (d) 16000 m

Answer: (a)

Question: In a transformer if primary and secondary coil have 100 and 10 turns and primary voltage is 220V. If the secondary coil has 2 resistors 7Ω and 15Ω as shown in the figure, find the potential difference across 7Ω resistance.



Options:

- (a) 15 V
- (b) 7 V
- (c) 22 V
- (d) 11 V

Answer: (b)

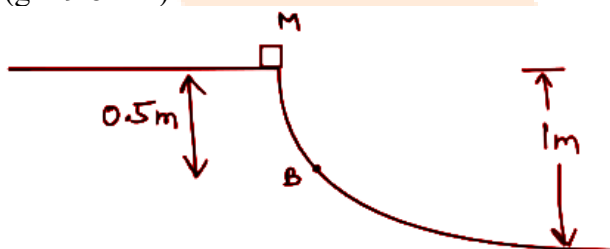
Question: The distance between an object and its twice magnified real image for a convex lens is 45 cm. Find the focal length of the lens

Options:

- (a) 15 cm
- (b) 10 cm
- (c) 30 cm
- (d) 60 cm

Answer: (b)

Question: If a block of mass 'M' is released from the top of a frictionless slide, find the velocity when the block reaches to point 'B' that 0.5 m below the starting point. ($g = 9.8 \text{ m/s}^2$)

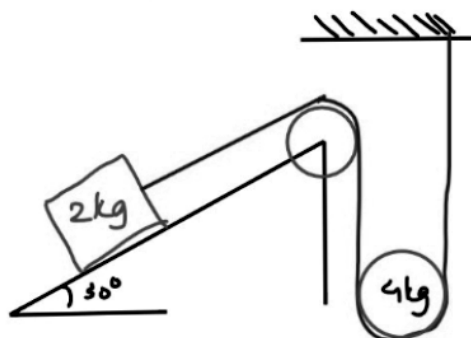


Options:

- (a) 3.14
- (b) 6.28
- (c) 3.4
- (d) 4.2

Answer: (a)

Question: Find the acceleration of 2 kg block. (Surface of inclined plane is smooth)



Options:

- (a) g
- (b) $g/2$
- (c) $g/3$
- (d) $g/4$

Answer: (c)

Question: Find the temperature at which RMS speed of H_2 will be same as that of RMS speed of Oxygen at $47^\circ C$

Options:

- (a) $20^\circ C$
- (b) $-253^\circ C$
- (c) $0^\circ C$
- (d) $-253 K$

Answer: (b)

Question: A disc of radius 2m and mass 5 kg is rotating about its vertical axis at 10 rad/sec. if another identical disc at rest is kept on top of it coaxially, then find the energy dissipated by the time slipping stops

Options:

- (a) 500 J
- (b) 1000 J
- (c) 250 J
- (d) 100 J

Answer: (c)

Question: Two identical insulated coil carrying same current I (same radius a) & placed such that their centers are coinciding with each other, but planes of coil are perpendicular to each other. Find net magnetic induction at the center due this arrangement.

Options:

- (a) $\frac{\sqrt{2}\mu_0 I}{4a}$
- (b) $\frac{\mu_0 I}{\sqrt{2}a}$
- (c) $\frac{\sqrt{2}\mu_0 I}{a}$
- (d) $\frac{\sqrt{3}\mu_0 I}{2a}$

Answer: (b)

Question: If electric field component of an EM wave is

$$\vec{E} = E_0 \cos(\omega t - kz) \hat{i} \text{ then the magnetic component will}$$

Options:

(a) $\vec{B} = E_0 C \cos(\omega t - kz) \hat{j}$

(b) $\vec{B} = \left(\frac{E_0}{C} \right) \cos(\omega t - kz) \hat{j}$

(c) $\vec{B} = (E_0 C) \cos(\omega t - kz) (-\hat{j})$

(d) $\vec{B} = \left(\frac{E_0}{C} \right) \cos(\omega t - kz) (-\hat{j})$

Answer: (b)

Question: In a closed organ pipe, fundamental frequency f is 50 Hertz. Now same water is filled and frequency becomes 110 Hz. if the cross sectional area of the pipe is 2 cm^2 , then find the amount of water added in grams. Speed of sound in air = 330 m/s.

Options:

(a) 90 grams

(b) 180 grams

(c) 300 grams

(d) 18 grams

Answer: (b)

Question: For uniform accelerated motion a body travels 125m from t to $t+1$ s while increasing speed by 50 m/s, then the displacement in the next second from $t+1$ to $t+2$ is ?

Options:

(a) 175 m

(b) 165 m

(c) 186 m

(d) 195 m

Answer: (a)

Question: Dimensions of viscosity coefficient, surface tension, angular momentum, rotational energy ?

Options:

(a) $M^1 L^{-1} T^{-1}$

(b) MT^{-2}

(c) $M^1 L^2 T^{-1}$

(d) $M^1 L^2 T^{-2}$

Answer: (a)

Question: Electrostatic potential due to a short dipole is proportional to

Options:

(a) r

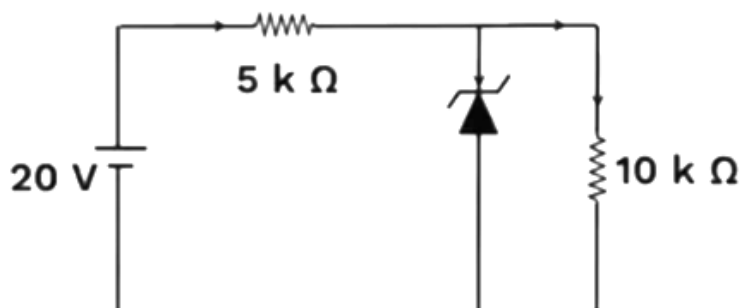
(b) $1/r$

(c) $1/r^2$

(d) $1/r^3$

Answer: (c)

Question: Breakdown voltage of zener diode is 10 V then the current through it is ?



Options:

- (a) 1 mA
- (b) 2 mA
- (c) 3 mA
- (d) 4 mA

Answer: (a)

Question: If the power factor of LR circuit having voltage $E = 25 \sin(1000t)$ is $1/\sqrt{2}$ then the power factor of $E = 20 \sin(2000t)$ is ?

Options:

- (a) $1/\sqrt{5}$
- (b) $1/\sqrt{3}$
- (c) $1/2$
- (d) $1/\sqrt{7}$

Answer: (a)

Question: A ball of 100 g is dropped from a height of 10 m and after collision with ground rises to 5 m. Find the impulse by the ground surface ?

Options:

- (a) 0.09 Ns
- (b) 0.28 Ns
- (c) 0.38 Ns
- (d) 0.29 Ns

Answer: (d)

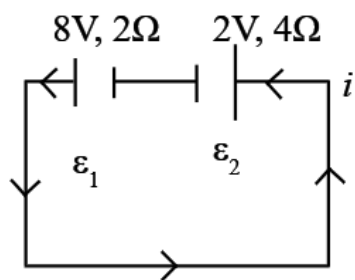
Question: A wire carrying $\sqrt{2}$ A current is placed in the plane of magnetic field of 3.5×10^{-5} T making 45° . Find the force per unit length on it ?

Options:

- (a) 34×10^{-6} N/m
- (b) 34×10^{-5} N/m
- (c) 35×10^{-6} N/m
- (d) 35×10^{-5} N/m

Answer: (c)

Question:



Voltage drop across ϵ_2 ?

Options:

- (a) 6 V
- (b) 5 V
- (c) 4 V
- (d) 7 V

Answer: (a)



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Chemistry

Question: Statement I: $\text{CH}_2 = \text{CH}_2 - \text{CH}_2 - \text{Cl}$ is an allyl halide

Statement II: Allyl halides have halogen attached to sp^2 carbon

Options:

- (a) Both statement I and statement II are false
- (b) Statement I is true but statement II is false
- (c) Statement I is false but statement II is true
- (d) Both statement I and statement II are true

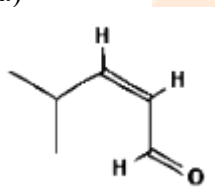
Answer: (b)

Solution: Statement I correct and II is wrong

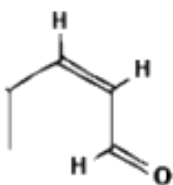
Question: Find the correct structure of 4-methylpent-2-enal

Options:

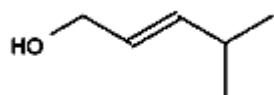
(a)



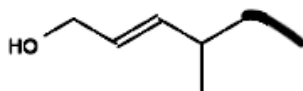
(b)



(c)

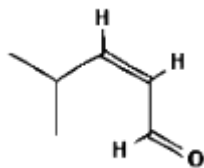


(d)



Answer: (a)

Solution:



Question: Which of the following lanthanides ions are diamagnetic?

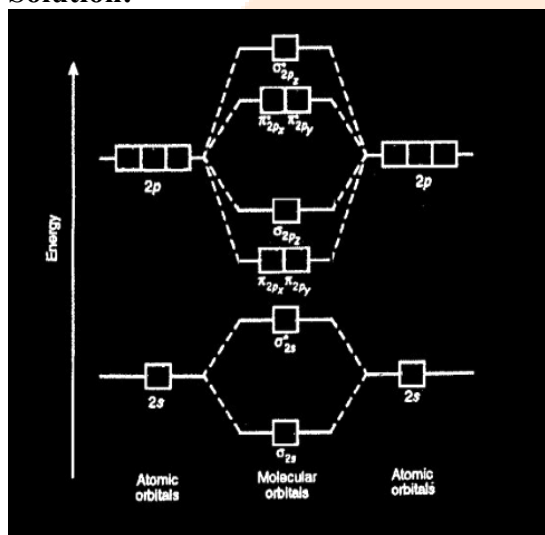
Options:

- (a) Ce^{+4} & La^{+3}
- (b) Ce^{+4} & Sm^{+2}
- (c) Ce^{+2} & Yb^{+4}
- (d) Ce^{+2} & Yb^{+2}

Answer: (a)

Question: Find out the maximum number of hybrid orbitals formed when 2s and 2p orbitals are mixed.

Solution:



Question: IUPAC name Unununium element lies in which group in the periodic table?

Options:

- (a) 7th period., 11 group
- (b) 5th group
- (c) 6th Group
- (d) 4th group

Answer: (a)

Solution:

Periodic table with atomic number, symbol, and electron configuration

group	1*	2											13	14	15	16	17	18	
1	1	2											3	4	5	6	7	8	9
2	3	4											5	6	7	8	9	10	11
3	11	12											13	14	15	16	17	18	19
4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55
6	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73
7	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105

☐ Alkali metals ☐ Halogens
☐ Alkaline-earth metals ☐ Noble gases
☐ Transition metals ☐ Rare-earth elements (21, 39, 57–71) and lanthanoid elements (57–71 only)
☐ Other metals ☐ Actinoid elements
☐ Other nonmetals

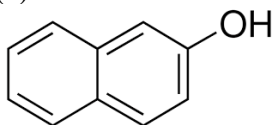
lanthanoid series 6

actinoid series 7

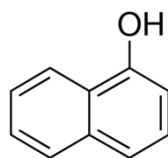
Question: Diazonium salt reacts with X compound to give the scarlet red colour. What is X?

Options:

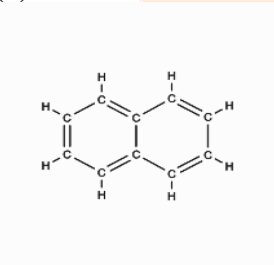
(a)



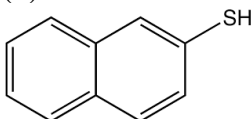
(b)



(c)

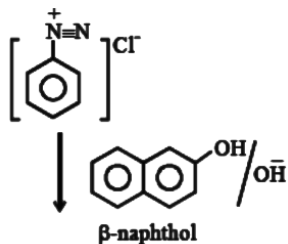


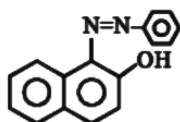
(d)



Answer: (a)

Solution:





Orange-Red. dye

Question: Statement I: Same energy level orbitals are called degenerate orbitals

Statement II: In hydrogen the 3p and 3d orbitals are not degenerate orbitals

Options:

- (a) Both statement I and statement II are false
- (b) Statement I is true but statement II is false
- (c) Statement I is false but statement II is true
- (d) Both statement I and statement II are true

Answer: (b)

Solution: Statement: 1 is true and

Statement: 2 is false as there are no electrons in the third orbit and thus 3s and 3p and 3d are of equal energy.

Question:



Options:

- (a) H_2 / Pd - BaSO_4
- (b) LiAlH_4
- (c) NH_2NH_2 / KOH / CH_2OH - CH_2OH
- (d) Zn - Hg/HCl

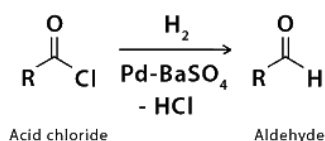
Answer: (a)

Solution:

$\text{Pb BaSO}_4 + \text{H}_2$

lindlar's catalyst its rosenmund reaction.

Rosenmund Reduction



Question: Which of the following will not give Fehling test

Options:

- (a) Acetone
- (b) Propanal
- (c) Ethanal
- (d) Butanal

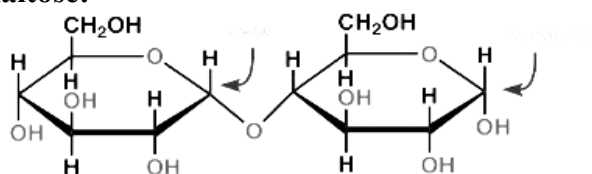
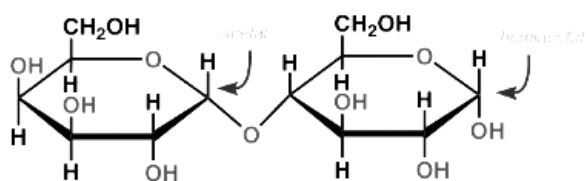
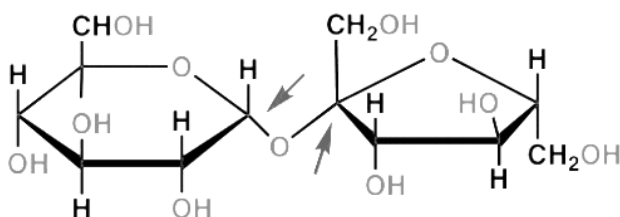
Answer: (a)

Question: Assertion: Maltose and lactose are examples of reducing sugars.

Reason: Maltose and lactose reducing Fehling's solution and Tollen's reagent

Options:

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
 (c) Assertion is correct but Reason is incorrect
 (d) Both Assertion and Reason are incorrect

Answer: (a)**Solution:** Lactose and Maltose are also reducing sugars and give a positive Benedict test**Maltose:****Lactose:****Sucrose is a non-reducing sugar**

The structure lacks any hemiacetal functional groups and is therefore “locked” in its cyclic form

Question: What is mass of sodium acetate so as to create. A 250 ml of 0.35 M solution.
 Molar mass was given 82.02 gn

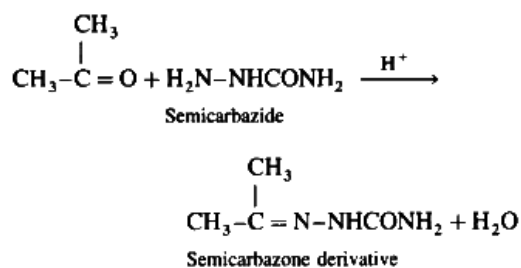
Solution:

$$0.35 \times 82 \times 0.25 \Rightarrow \text{Mass} \\ \Rightarrow 7.175$$

Question: If ethanol is reacted with semicarbazide. How many nitrogen atom In product

Solution:

3 Nitrogen in product



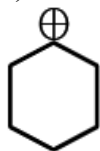
Question: Which of the following compound or ion is most stable ?

Options:

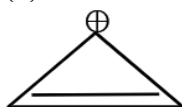
(a)



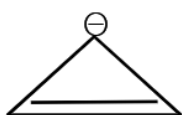
(b)



(c)



(d)



Answer: (c)

Solution: Due to aromaticity

Question: What is the geometry of Aluminium chloride in aqueous solution

Options:

(a) Square planar

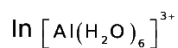
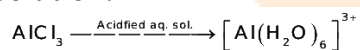
(b) Octahedral

(c) Tetrahedral

(d) Square pyramidal

Answer: (b)

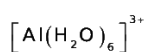
Solution:



Oxidation state of $\text{Al}^{+3} : 1s^2 2s^2 2p^6$

$=[\text{Ne}]$

Hybridisation $= sp^3d^2$



hybridisation - sp^3d^2

Question: Choose the correct option

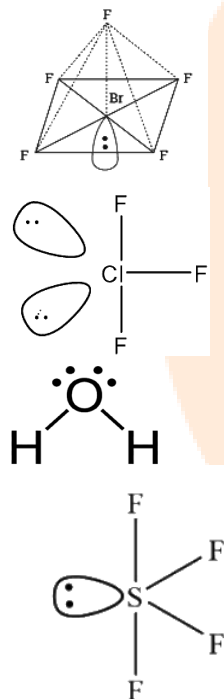
Molecule	Shape
(a) BrF_5	(i) Sea-saw
(b) H_2O	(ii) T-shape
(c) ClF_3	(iii) Bent
(d) SF_4	(iv) Square Pyramid

Options:

- (a) (A) -iv; (B) - iii; (C) - ii; (D) - i
 (b) (A) -iv; (B) - iii; (C) - i; (D) - ii
 (c) (A) -iii; (B) - iv; (C) - ii; (D) - i
 (d) (A) -iii; (B) - iv; (C) - i; (D) - ii

Answer: (a)

Solution:



See-saw shape

Question: What will happen to freezing point if we add naphthalene in benzene

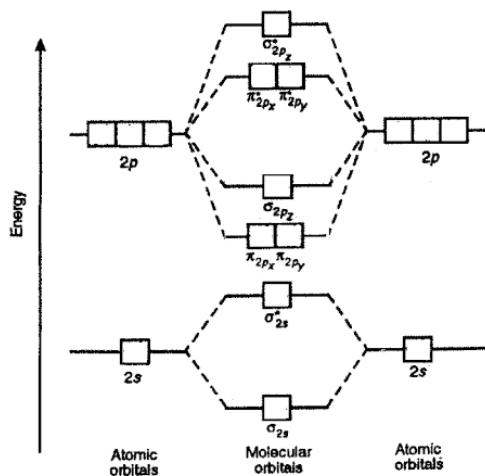
Options:

- (a) Increases
 (b) Decreases
 (c) First Increases then decreases
 (d) Remains same

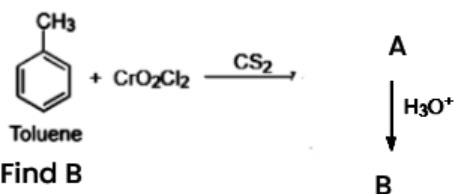
Answer: (b)

Question: Find out the maximum number of molecular orbitals formed when 2s and 2p orbitals are mixed in diamagnetic molecule

Solution:



Question:



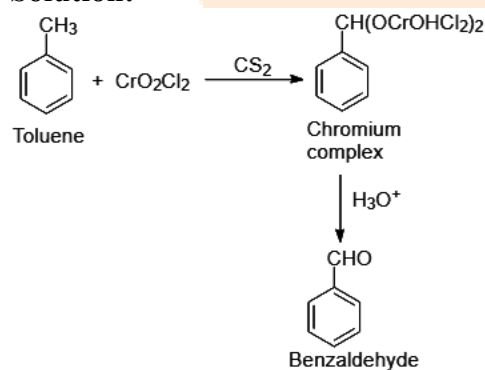
Find B

Options:

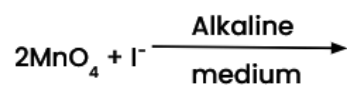
- (a) Acetone
- (b) Propanal
- (c) Benzaldehyde
- (d) Butanal

Answer: (c)

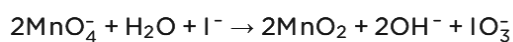
Solution:



Question: Find out sum of the coefficients of all the species involved in the balanced equation :



Solution:



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[MORNING SHIFT]

Maths

Question: Two circles having eqn $(y+2)^2 + (x+1)^2 = r^2$

And $y^2 + x^2 - 4x - 4y + 4$

Solution:

Circles: $(x+1)^2 + (y+2)^2 = r^2$, $x^2 + y^2 - 4x - 4y + 4 = 0$

Centres = $(-1, -2)$, $(2, 2)$

radil = $r, 4$

$$d = \sqrt{9+16} = 5$$

$$\Rightarrow 1r - 4 < 5 < r + 4$$

$$\Rightarrow r > 1, -5 < r - 4 < 5 \Rightarrow -1 < r < 9$$

$$\Rightarrow r \in (1, 9)$$

OR $r \in (-9, -1) \cup (1, 9)$ if r is not given +ve.

Question: Area bounded by $y^2 = 4(x - 2)$ and $y = 2x - 8$

Solution:

Parabola: $y^2 = 4(x - 2)$, line : $y = 2x - 8$

$$\text{Solving : } 4(x - 4)^2 = 4(x - 2)$$

$$\Rightarrow x^2 - 9x + 18 = 0 \Rightarrow x = 3, 6$$

$$\text{Area} = \frac{d^3}{6l} = \frac{6^3}{6 \times 4} = 9$$

Question: In maths, $S_{20} = 790$, $S_{10} = 145$, find $S_{15} - S_5 \dots$ sum of terms of ap

Solution:

$$AP : S_{20} = 790$$

$$\Rightarrow \frac{20}{2}(2A + 19d) = 790 \Rightarrow 2a + 19d = 79$$

$$S_{10} = 145 \Rightarrow \frac{10}{2}(2a + 9d) = 145 \Rightarrow 2a + 9d = 29$$

$$\Rightarrow 10d = 50 \Rightarrow d = 5; 2a = -16 \Rightarrow A = -8$$

$$S_{15} - S_5 = \frac{15}{2}(2A + 14d) - \frac{5}{2}(2a + 4d)$$

$$= \frac{5}{2}(6a + 42d - 2a - 4d)$$

$$= \frac{5}{2}(4a + 38d) = 5(2a + 19d)$$

$$= 5 \times 79 = 395$$

Question: $f(x) = \frac{1}{|x|}$ if $|x| < 2$, $-ax^2 + 2b$. Find $48(a + b)$

Solution:

$$f(x) = \begin{cases} \frac{1}{|x|} & |x| < 2 \\ -ax^2 + 2b & |x| \geq 2 \end{cases}$$

$$\frac{1}{2} = -4a + 2b \quad \frac{1}{4} = -2a + b \quad \dots(1)$$

$$f(x) = \begin{cases} -\frac{1}{x^2}, & 0 < x < 2 \\ -2ax, & x > 2 \end{cases}$$

$$-\frac{1}{4} = -4a \Rightarrow a = \frac{1}{16}, \frac{1}{4} = \frac{-1}{8} + 6$$

$$b = \frac{3}{8}$$

$$48(a + b) = 48a + 48b = 3 + 18 = 21$$

Question: If the length of the minor axis of an ellipse is equal to half of the distance between the foci, then the eccentricity of the ellipse is.

Solution:

$$2b = \frac{1}{2} \cdot 2ae \Rightarrow 2b = ae$$

$$\Rightarrow 4b^2 = a^2 e^2$$

$$\Rightarrow 4a^2(1 - e^2) = a^2 e^2$$

$$\Rightarrow 4 - 4e^2 = e^2$$

$$\Rightarrow e = \frac{2}{\sqrt{5}}$$

Question: Domain of $\cos^{-1}\left(1 - \frac{|x|}{4}\right) + \{\log(3-x)^{-1}\}$

Solution:

$$f(x) = \cos^{-1}\left(1 - \frac{|x|}{4}\right) - 1 \frac{1}{\log(3-x)}$$

$$\Rightarrow -1 \leq 1 - \frac{|x|}{4} \leq 1, 3-x > 0, 3-x \neq 1$$

$$\Rightarrow -2 \leq -\frac{|x|}{4} \leq 0, x < 3, x \neq 2$$

$$\Rightarrow 8 \geq |x| \geq 0$$

$$\Rightarrow x \in [-8, 3) - \{2\}$$

$$a + b + c = 8 + 3 + 2 = -3$$

Question: Number of integral values of $\left\{7^{\frac{1}{2}} + 11^{\frac{1}{6}}\right\}^{824}$

Solution:

$$\left(7^{\frac{1}{2}} + 11^{\frac{1}{6}}\right)^{824}$$

$$T_{K+1} = {}^{824}C_k \left(7^{\frac{1}{2}}\right)^{824-k} \left(11^{\frac{1}{6}}\right)^k$$

$$\frac{824 \cdot k}{2} \text{ and } \frac{k}{6} \in \mathbb{Z}$$

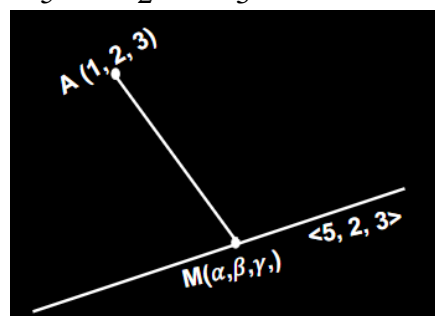
$$k = 6m$$

$$k = \left\lceil \frac{275}{6} \right\rceil + 1$$

$$k = 138$$

Question: Let (α, β, γ) be the foot of perpendicular from the point $(1, 2, 3)$ on the line

$$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3} \text{ then } 19(\alpha + \beta + \gamma)$$



Solution:

Let $\alpha = 5\lambda - 3$

$\beta = 2\lambda + 1$

$\gamma = 3\lambda - 4$

DR's of AM:

$< 5\lambda - 4, 2\lambda - 1, 3\lambda - 7 >$

$\therefore 25\lambda - 20 + 4\lambda - 7 + 9\lambda - 21 = 0$

$38\lambda = 43$

$\lambda = \frac{43}{38}$

$19(\alpha + \beta + \gamma) = 19[10\lambda - 6]$

$= 215 - 114 = 101$

Question: If $z = x + iy$, $xy \neq 0$ satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to

Solution:

$z^2 + i\bar{z} = 0$

$\rightarrow |z|^2 = |z|$

$\rightarrow |z| = 0$ or

Rejected.

Question: The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n^3}{(n^2 + k^2)(n^2 + 3k^2)}$

Solution:

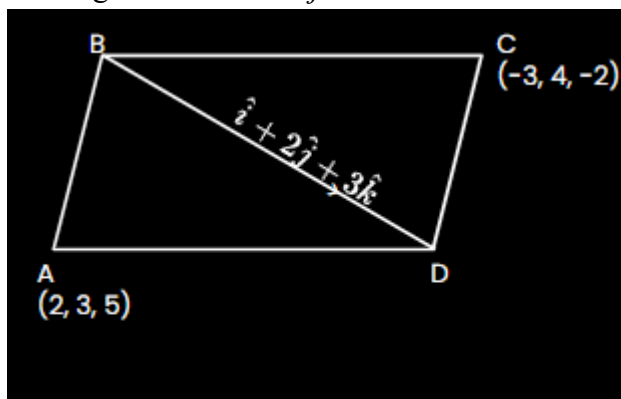
$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\frac{1}{n}}{\left(1 + \frac{k^2}{n^2}\right) \left(1 + \frac{3k^2}{n^2}\right)}$$

$$= \int_0^1 \frac{dx}{(1+x^2)(1+3x^2)} = \frac{1}{3} \int_0^1 \frac{(x^2+1) \left(x^2 + \frac{1}{3}\right)}{(1+x^2) \left(x^2 + \frac{1}{3}\right)} dx \cdot \frac{3}{2}$$

$$= \frac{1}{2} \left[\frac{1}{53} (\tan^{-1} 53x)_0^1 - (\tan^{-1} x)_0^1 \right]$$

$$= \frac{1}{2} \left(53 \cdot \frac{\pi}{3} - \frac{\pi}{4} \right) = \frac{\pi}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{4} \right)$$

Question: Let $A(2, 3, 5)$ and $C(-3, 4, -2)$ be 'opposite vertices of a parallelogram ABCD. If the diagonal $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$ then the area of the parallelogram is equal to



Solution:

$$\overrightarrow{AC} = (-5\hat{i} + \hat{j} - 7\hat{k})$$

$$\text{Area} = \frac{1}{2} |\overrightarrow{AC} \times \overrightarrow{BD}|$$

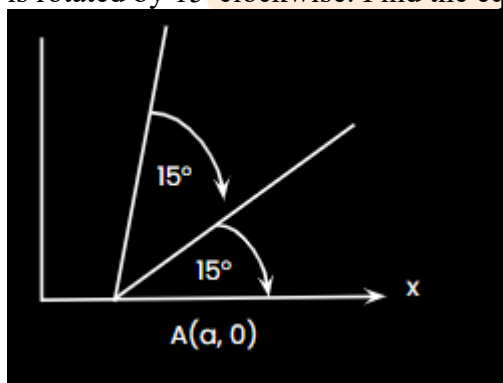
$$= \frac{1}{2} \begin{vmatrix} 75 & -24 \\ -24 & 14 \end{vmatrix}$$

$$= \frac{1}{2} [75 \times 14 - 24^2]$$

$$= \frac{1}{2} [1050 - 576]$$

$$= \frac{\sqrt{474}}{2}$$

Question: If a straight line passes through $A(a, 0)$ & makes an angle 30° with the x-axis. If it is rotated by 15° clockwise. Find the equation of new straight line?

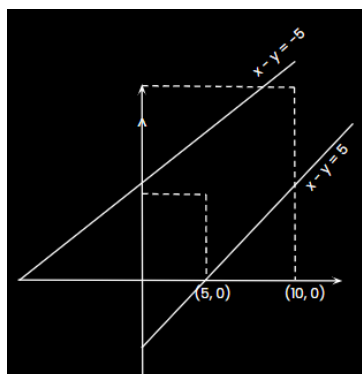


Solution:

$$y - 0 = \tan 15^\circ (x - a)$$

$$y = (2 - \sqrt{3})(x - a)$$

Question: Find probability that $|x-y| > 5$ given x, y belong to $\{0, 1, 2, \dots, 10\}$



Solution:

$$x, y \in \{0, 1, 2, \dots, 10\}$$

$$\rho = \frac{(5+4+3+2+1) \times 2}{11 \times 11}$$

$$= \frac{30}{121}$$

Question: If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3 + \sin^2 2x \\ 3 + 2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3 + 2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to

Solution:

$$f'(0) = \begin{vmatrix} 0 & . & . \\ 0 & . & . \\ 0 & . & . \end{vmatrix} + \begin{vmatrix} 2 & 0 & 3 \\ 5 & 0 & 0 \\ 2 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 3 & 0 \end{vmatrix}$$

$$= 0$$

Question: $g(x)$ is non constant twice differentiable function

Options:

(a) $f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 1$

(b) $f'(x) = 0$ for at least 1 value of $x \in (0, 2)$

(c) $f'(x) = 0$ for no. value of $x \in (0, 1)$

(d) $f'(x) = 0$ for exactly one value of $x \in (0, 1)$

Question: Median $\rightarrow M$

$$20M = ?$$

Class	0-4	4-8	8-12	12-16	16-20
Frequency	3	6	10	12	5

Solution:

$$M = L + \left[\frac{\frac{n}{2} - c}{f} \right] \times h$$

$$n = 36, L = 8, c = 9, f = 10, h = 4$$

$$M = 8 + \frac{18 - 9}{10} \times 4$$

$$= 8 + 3 \cdot 6 = 11.6$$

Class	Frequency	c.l
0-4	3	3
4-8	6	9
8-12	10	19
12-16	12	31
16-20	5	36

Median class